

Himanshu Ranjan

413-265-4410 | himanshuranj@umass.edu | [LinkedIn](#) | [GitHub](#)

Education

University of Massachusetts Amherst

Amherst, MA

Master of Science in Computer Science (GPA: 3.95/4.0)

Aug. 2025 – May 2027

Indian Institute of Technology Kharagpur

Kharagpur, India

Bachelor of Technology in Electrical Engineering (GPA: 8.22/10.0)

Aug. 2018 – May 2022

Experience

Persona AI

Houston, TX

Autonomy / World Modeling Intern

Sep. 2026 – Dec. 2026

- Developing **world-modeling and autonomy infrastructure for humanoid robots**, including model post-training, environment understanding, and complex industrial manipulation tasks.
- Building **NVIDIA Isaac Sim and NVIDIA Cosmos 3 workflows** to evaluate perception, prediction, task reasoning, robot-skill integration, and safety-critical autonomy.

Embodied AGI Lab, UMass Amherst

Amherst, MA

*Student Researcher; Advisor: Prof. Chuang Gan; Ph.D. Mentor: Junyi Cao
submitted to ICLR 2027*

Jul. 2026 – Present

- Built a **3D Gaussian Splatting digital twin** in the Genesis simulator for real-robot manipulation, using privileged state and task signals to generate dense and sparse supervision for vision-based critics.
- Trained and evaluated **SmolVLA, DiT diffusion, Pi0, and Pi0.5** manipulation policies across single-arm xArm7 and bimanual RM65B platforms on 8 real-robot manipulation tasks.
- Evaluated digital-twin-derived critic and reward signals for offline post-training against sparse-reward and learned reward-model baselines, measuring policy gains and **sim-to-real transfer**.

Human-Centered Robotics Lab, UMass Amherst

Amherst, MA

Student Researcher; Advisor: Prof. Hao Zhang; Ph.D. Mentor: Oscar Youngquist

Jun. 2026 – Present

- Built a real-robot data pipeline for the **Unitree Go2**, time-aligning RGB-D, GPS, IMU, and teleoperation commands in ROS 2 for multimodal dataset generation and offline evaluation.
- Developed a **classifier-guided multimodal navigation policy** using temporal sensor encoders, a context transformer, and diffusion and flow-matching models to predict future motion commands.
- Deployed the policy in a **distributed ROS 2 inference stack** with asynchronous sensor encoders, 5 Hz policy control, freshness checks, telemetry, and guarded controller handoff.

Bosch Global Software Technologies

Bangalore, India

Machine Learning Engineer / Senior Software Engineer

Aug. 2022 – Jul. 2025

- Designed a trifocal detection head that fused multi-range camera views, reducing runtime by **35%** and improving multi-task stability through task-specific filtering and uncertainty weighting.
- Increased GPU utilization from **30% to 80%** by profiling the pole detection training pipeline and redesigning ground-truth preprocessing for more GPU-efficient multi-task training.
- Built an end-to-end pole detection pipeline with Python bindings for faster post-processing, Azure-based result storage, and **Voxel FiftyOne** visualization for prediction analysis and edge-case debugging.
- Deployed foundation models for image retrieval on **Triton Inference Server** for real-time inference.

LG Soft India

Bangalore, India

Research Intern

May 2021 – Jul. 2021

- Built a scene matching pipeline with **Mask R-CNN** segmentation and Siamese network similarity learning.
- Compared CNN and ViT-style embeddings for scene retrieval and false-match analysis.

Projects

Voice-Guided Person-Following Robot (Co-authored) | [GitHub](#)

May 2026

- Developed a **ROS 2-based robotics stack** for a RoboMaster EP, converting voice commands into structured robot intents for autonomous task execution.
- Built a target-following perception layer using person tracking, depth estimation, and face identity matching.
- Implemented FSM-gated navigation and manipulation, evaluating follow-distance using logged depth estimates.

Technical Skills

Languages: Python, C++, SQL, Bash

ML & AI: Deep Learning, Computer Vision, Reinforcement Learning, Imitation Learning, Diffusion Models, VLA Policies, LLMs, VLMs, Multimodal AI, World Models, Gaussian Splatting, NeRF, Sim-to-Real

Robotics & Simulation: ROS 2, LeRobot, NVIDIA Isaac Sim, Genesis, NVIDIA Cosmos 3

Frameworks & Tools: PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, Open3D, LangChain, LangGraph, ONNX, Triton Inference Server, Docker, Kubernetes, Git, Linux, Azure, AWS

Courses: ML, Reinforcement Learning, Computer Vision, Robotics, NLP, AI Alignment, Optimization Theory